The converse of the ρ-shift vanishing law

Q559: a complete vanishing criterion for the one-head $\Psi^{\mathrm{C}}_{(1^r),\theta}$

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Provenance (PROTOCOL §2.3).
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Repository: `clio-vega/proofs`. **Commit:** `02a264c` (this result); scripts and pre-registration at `303d72a`; provenance block added at the commit named in the covering email.
Registry: `proofs/registry/rick-q488-converse.json`, node `clio-Q553-vanishing-converse-Q559`, trust `proved`.

Abstract

For my one-head object $\Psi^{\mathrm{C}}_{(1^r),\theta}(M)$ over a finite set M of n indeterminates, I proved on 2026-10-22 that $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ whenever the ρ-shifts $m_j = \theta_j + j - 1$ have a repetition. Here I prove the converse, and therefore a complete criterion: for $n \geq r$, $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ in $\mathbb{Q}(w_1, \ldots, w_n)$ *if and only if* $m_j = m_k$ for some $j < k$ (Theorem 11, Corollary 12). The hypothesis $n \geq r$ is necessary and was missing from the question as I had posed it (Proposition 13).

The proof is a leading-term argument, but not in the grading I expected. It runs in the *single* variable w_n over $K = \mathbb{Q}(w_1, \ldots, w_{n-1})$: the subsets T containing the index n contribute order $w_n^{\max_j m_j - r + 1}$ with leading coefficient $\pm\Psi^{\mathrm{C}}_{(1^{r-1}),\hat{\theta}}(M \setminus \{w_n\})$ — a *rank-lowering branching rule* (Lemma 7) — while the subsets avoiding n contribute order at most w_n^{-r} (Lemma 5). The first block strictly dominates exactly when $\max_j m_j \geq 0$, which closes that case by induction on r. The complementary case is closed not by pushing the same argument but by the reflection $w \mapsto 1/w$, which acts on the ρ-shift by $m \mapsto -m$ (Lemma 9). The residual case $\max_j m_j = -1$ is a genuine *order tie* between the two blocks, measured to tie in 25/25 instances, and the leading-term instrument has no purchase there: the reflection is essential, not decorative.

Finally, non-vanishing is *generic and not universal*, with a structural witness: if M contains a reciprocal pair $w_a w_b = 1$ and $n \leq r + 1$, then every $B(T)$ vanishes and so $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ for *every* θ (Proposition 16). I found that witness by refusing to accept a negative control that read 58/60.

1 Pre-registration

Written before testing any predicate, and graded arm by arm in §7. Verbatim from `code-1023-q559/PREREG.md`, published with this note.

- **P1** (the brief's): *the extremal-monomial argument closes the general case.* Arms `CONFIRMED`/`SPLIT` (works for a restricted class of θ only — say which)/`REFUTED`. *Predicted:* `SPLIT`, for three reasons recorded in advance: (a) the brief's strategy-1 is ill-posed as literally stated, because the summands $B(T)\det/V$ are not polynomials in w — $B(T)$ carries the denominator

$\prod_{t\notin T}\prod_{x\in T}(1-w_t/x)$ — so "a monomial occurring in exactly one T-summand" has no meaning until a common denominator is cleared, and clearing it destroys the T-indexing; (b) I expected a leading-term argument to work instead in the grading by order at $w_n\to\infty$ over $K=\mathbb{Q}(w_1,\dots,w_{n-1})$; (c) I predicted a residual case $\max_j m_j\le -1$ that this argument does *not* close.

- **P2**: $B(T)$ genuinely takes both signs for some $r=2$ instance in the test range. *Predicted:* `TRUE`.
- **P3**: non-vanishing is generic, not universal. *Predicted:* `CONFIRMED`, with the reciprocal-pair witness already in hand, plus the structural prediction that the kill is total exactly when $n\le r+1$.
- **P4** (mine): the two-block leading-order lemma, stated in advance with its exponents and both signs. Arms `CONFIRMED`/wrong sign/wrong exponent/`REFUTED`.
- **P5** (mine): the reflection identity $\Psi^{\mathrm{C}}_{(1^r),\theta}(1/w)=(-1)^r e^{2r}\Psi^{\mathrm{C}}_{(1^r),\theta^\vee}(w)$ with $m^\vee=-m$. Arms `CONFIRMED`/`REFUTED`.
- **Omitted hypothesis**, flagged in advance: the conjecture needs $n\ge r$.

2 Setting

Convention 1. $M=\{w_1,\dots,w_n\}$ is a set of n independent indeterminates over $\mathbb{Q}$, carrying the index order $1<2<\cdots<n$. For $r\ge 0$ and $\theta\in\mathbb{Z}^r$ put $m_j=\theta_j+j-1$ $(1\le j\le r)$. Three different objects named Ψ are live in the surrounding conversation (Ikeda–Naruse's GKM Ψ_n, Rick's $\Psi_{\kappa,\theta}$, Rupel's Ψ_i); Ψ^{C} denotes *mine*, defined by the head-injection sum

$$\Psi^{\mathrm{C}}_{\kappa,\theta}(M)=\sum_{\substack{h=(h_1,\dots,h_r)\\ \text{distinct in } M}}\prod_{j=1}^{r}\bigl(1-X_j^{-2}\bigr)\rho(\kappa_j,X_j,w)X_j^{\theta_j}\prod_{j<k}\frac{1-\frac{1}{X_jX_k}}{1-\frac{X_j}{X_k}}\prod_{\substack{t\in M\\ t\notin h}}\prod_{j=1}^{r}\frac{1-\frac{1}{w_tX_j}}{1-\frac{w_t}{X_j}},\tag{1}$$

with $X_j=w_{h_j}$ and $\rho(\kappa,X,w)=\prod_{k<\kappa}(1-w_k/X)$. Throughout $\kappa=(1^r)$, so $\rho\equiv 1$. All displayed denominators are nonzero in $\mathbb{Q}(w_1,\dots,w_n)$, so $\Psi^{\mathrm{C}}_{(1^r),\theta}(M)$ is a well-defined element of that field. For $r=0$ the sum has the single empty term and $\Psi^{\mathrm{C}}_{(1^0),()}(M)=1$. Write $e=e(M)=\prod_{t=1}^{n}w_t$.

I take as given the alternant expansion of 2026-10-22, re-verified here (§8, 386/386) by agreement of two independent implementations.

Theorem 2 (alternant expansion; `clio-Q553-alternant-expansion`)**.** *For every finite M and every $\theta\in\mathbb{Z}^r$,*

$$\Psi^{\mathrm{C}}_{(1^r),\theta}(M)=\sum_{\substack{T\subseteq M\\ |T|=r}}B(T)\cdot\frac{\det\bigl(x_i^{m_j}\bigr)_{i,j=1}^{r}}{V(T)},\qquad V(T)=\prod_{1\le i<k\le r}(x_k-x_i),$$

where $x_1,\dots,x_r$ are the w-values of T in increasing index order and

$$B(T)=\prod_{x\in T}\bigl(1-x^{-2}\bigr)\cdot\prod_{\{x,y\}\subseteq T}\Bigl(1-\tfrac{1}{xy}\Bigr)\cdot\prod_{t\in M\setminus T}\prod_{x\in T}\frac{1-\frac{1}{w_tx}}{1-\frac{w_t}{x}}.\tag{2}$$

Theorem 3 (ρ-shift vanishing law, forward direction, 2026-10-22)**.** *If $m_j = m_k$ for some $j < k$ — equivalently $\theta_j - \theta_k = k - j$ — then $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$.*

The matrix $(x_i^{m_j})$ then has two equal columns, so every summand of Theorem 2 vanishes. The question of this note is the converse.

2.1 Why it is not free

When the m_j are pairwise distinct every summand is a nonzero $\pm B(T)s_\mu(T)$ times a monomial, so no individual term vanishes; but $B(T)$ is *not* of one sign (P2, §7: both signs occur in 400/400 sampled $r = 2$ instances), so there is no positivity argument available and the signed sum could in principle cancel identically.

3 The order filtration at $w_n \to \infty$

Convention 4. Let $K = \mathbb{Q}(w_1, \dots, w_{n-1})$ and regard $\Psi^{\mathrm{C}}_{(1^r),\theta}(M)$ as an element of $K(w_n)$, embedded in the field of formal Laurent series $K((w_n^{-1}))$ by expanding every rational function of w_n at $w_n = \infty$. For $0 \neq f = \sum_{k \geq k_0} c_k w_n^{-k}$ with $c_{k_0} \neq 0$ set $\deg_\infty f = -k_0$ and $\mathrm{lc}_\infty f = c_{k_0} \in K$; set $\deg_\infty 0 = -\infty$. Then $\deg_\infty(f+g) \leq \max(\deg_\infty f, \deg_\infty g)$, with equality and additive leading coefficients when $\deg_\infty f \neq \deg_\infty g$. Write $\mathcal{O} = 1 + w_n^{-1}K[[w_n^{-1}]]$, a multiplicative group. Everything below is an identity in $K((w_n^{-1}))$; no genericity or specialisation of $w_1, \dots, w_{n-1}$ is used anywhere.

Split Theorem 2 by whether T contains the largest index:

$$\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = \Psi^{\mathrm{C},+} + \Psi^{\mathrm{C},-}, \qquad \Psi^{\mathrm{C},\pm} = \sum_{\substack{|T|=r \\ n\in T \ (+),\ n\notin T \ (-)}} B(T)\frac{\det(x_i^{m_j})}{V(T)}.$$

Throughout, $M' = M \setminus \{w_n\} = \{w_1, \dots, w_{n-1}\}$, and B_M, $B_{M'}$ denote (2) formed over M and over M' respectively.

Lemma 5 (the block avoiding n)**.** $\deg_\infty \Psi^{\mathrm{C},-} \leq -r$. *Moreover the coefficient of w_n^{-r} in $\Psi^{\mathrm{C},-}$ equals $(-1)^r\,\Psi^{\mathrm{C}}_{(1^r),\theta+\mathbf{1}}(M')$, where $\theta + \mathbf{1} = (\theta_1 + 1, \dots, \theta_r + 1)$.*

Proof. Let $T \subseteq M'$ with $|T| = r$. Comparing (2) over M and over M', the only difference is the factor contributed by $t = n \in M \setminus T$:

$$B_M(T) = B_{M'}(T) \prod_{x\in T} \frac{1 - \frac{1}{w_n x}}{1 - \frac{w_n}{x}}.$$

For each $x \in T$, writing $1 - \frac{w_n}{x} = -\frac{w_n}{x}\big(1 - \frac{x}{w_n}\big)$,

$$\frac{1 - \frac{1}{w_n x}}{1 - \frac{w_n}{x}} = -\frac{x}{w_n} \cdot \frac{1 - \frac{1}{w_n x}}{1 - \frac{x}{w_n}} \in -\frac{x}{w_n}\,\mathcal{O},$$

since both $1 - \frac{1}{w_n x}$ and $\big(1 - \frac{x}{w_n}\big)^{-1}$ lie in $\mathcal{O}$. Multiplying the r factors, $B_M(T) \in (-1)^r w_n^{-r}\big(\prod_{x\in T} x\big) B_{M'}(T)\,\mathcal{O}$. The remaining factor $\det(x_i^{m_j})/V(T)$ lies in K, being independent of w_n. Hence

$$B_M(T)\frac{\det(x_i^{m_j})}{V(T)} = w_n^{-r}\Big((-1)^r\Big(\prod_{x\in T} x\Big)B_{M'}(T)\frac{\det(x_i^{m_j})}{V(T)} + O(w_n^{-1})\Big),$$

which gives $\deg_\infty \le -r$ for each T, hence for the finite sum $\Psi^{\mathrm{C},-}$. For the coefficient, $\big(\prod_{x\in T} x\big)\det(x_i^{m_j}) = \det(x_i^{m_j+1})$ by scaling each row, and $m_j+1 = (\theta_j+1)+j-1$ is the ρ-shift of $\theta+\mathbf{1}$; summing over $T\subseteq M'$ and applying Theorem 2 over M' identifies the coefficient as $(-1)^r\Psi^{\mathrm{C}}_{(1^r),\theta+\mathbf{1}}(M')$. □

Remark 6. Only the inequality $\deg_\infty \Psi^{\mathrm{C},-} \le -r$ is used in the proof of Theorem 11. The exact coefficient is recorded because it is independently measurable, and it is measured (137/137, §8): an identity I rely on and an identity I merely believe should not be verified by the same instrument, and this one carries no load.

Lemma 7 (the block containing n: a rank-lowering branching rule)**.** *Assume $m_1,\dots,m_r$ are pairwise distinct ($r\ge 1$) and let j^* be the unique index with $m_{j^*} = \max_j m_j$. Let $\hat m = (m_j)_{j\ne j^*}$ listed in increasing order of j, and $\hat\theta_l = \hat m_l - l + 1$ for $1\le l\le r-1$. Then*

$$\Psi^{\mathrm{C},+} = w_n^{m_{j^*}-r+1}\Big((-1)^{r+j^*}\,\Psi^{\mathrm{C}}_{(1^{r-1}),\hat\theta}(M') + O(w_n^{-1})\Big).$$

In particular $\deg_\infty \Psi^{\mathrm{C},+} \le m_{j^*}-r+1$, with equality iff $\Psi^{\mathrm{C}}_{(1^{r-1}),\hat\theta}(M')\ne 0$.

Proof. A subset $T\ni n$ with $|T|=r$ is $T = T'\sqcup\{n\}$ with $T'\subseteq M'$, $|T'| = r-1$. In the index order of T, the values are $x_1,\dots,x_{r-1}$ (those of T', in increasing index order) and $x_r = w_n$, since n is the largest index of M. Three computations.

(i) The weight factors. Separating in (2) the factors that involve w_n,

$$B_M(T'\sqcup\{n\}) = B_{M'}(T')\cdot\beta_{T'},\qquad \beta_{T'} = \big(1-w_n^{-2}\big)\prod_{x\in T'}\Big(1-\tfrac{1}{xw_n}\Big)\prod_{t\in M'\setminus T'}\frac{1-\frac{1}{w_t w_n}}{1-\frac{w_t}{w_n}}.$$

Indeed $\prod_{x\in T}(1-x^{-2})$ splits off $(1-w_n^{-2})$; the pair product $\prod_{\{x,y\}\subseteq T}(1-\frac{1}{xy})$ splits off $\prod_{x\in T'}(1-\frac{1}{xw_n})$; and $M\setminus T = M'\setminus T'$, so for each $t\in M'\setminus T'$ the inner product over $x\in T$ splits off the single factor with $x=w_n$. The surviving T'-only factors are exactly $B_{M'}(T')$. Each factor of $\beta_{T'}$ lies in $\mathcal{O}$, and $\mathcal{O}$ is a group, so $\beta_{T'}\in\mathcal{O}$.

(ii) The Vandermonde. $V(T) = V(T')\cdot\prod_{i=1}^{r-1}(w_n - x_i) = V(T')\,w_n^{r-1}\gamma_{T'}$ with $\gamma_{T'}\in\mathcal{O}$.

(iii) The alternant. Laplace expansion along the last row gives $\det(x_i^{m_j})_{i,j=1}^r = \sum_{j=1}^r(-1)^{r+j}w_n^{m_j}D_j$ with $D_j = \det(x_i^{m_{j'}})_{1\le i\le r-1,\ j'\ne j}\in K$. Because the m_j are pairwise distinct, j^* is the *unique* maximiser, so $m_j - m_{j^*}\le -1$ for all $j\ne j^*$ and

$$\det(x_i^{m_j})_{i,j=1}^r = w_n^{m_{j^*}}\Big((-1)^{r+j^*}D_{j^*} + O(w_n^{-1})\Big).$$

This is the one place the distinctness hypothesis enters, and it enters as the *strictness* of the maximum.

Combining (i)–(iii),

$$B_M(T)\frac{\det(x_i^{m_j})}{V(T)} = w_n^{m_{j^*}-r+1}\Big((-1)^{r+j^*}B_{M'}(T')\frac{D_{j^*}}{V(T')} + O(w_n^{-1})\Big).$$

Finally $D_{j^*} = \det\big(x_i^{\hat m_l}\big)_{i,l=1}^{r-1}$, so $D_{j^*}/V(T')$ is precisely the alternant ratio of Theorem 2 for the ρ-shift $\hat m$, i.e. for $\hat\theta$, over the subset T' of M'. Summing over the finitely many T' and applying Theorem 2 in rank $r-1$ over M' gives the claim. (For $r=1$ the sum has the single term $T'=\emptyset$ and $\Psi^{\mathrm{C}}_{(1^0),()}(M') = 1$.) □

Proposition 8 (the dominant case)**.** *Let $r \geq 1$, $n \geq r$, let m be pairwise distinct, and suppose*

$$\max_j m_j \;\geq\; 0 \qquad \textit{and} \qquad \Psi^{\mathrm{C}}_{(1^{r-1}),\hat{\theta}}(M') \neq 0.$$

Then $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) \neq 0$; indeed $\deg_\infty \Psi^{\mathrm{C}}_{(1^r),\theta}(M) = m_{j^*} - r + 1$ with leading coefficient $(-1)^{r+j^*}\Psi^{\mathrm{C}}_{(1^{r-1}),\hat{\theta}}(M')$.

Proof. By Lemma 7, $\deg_\infty \Psi^{\mathrm{C},+} = m_{j^*} - r + 1$ with the stated leading coefficient. Since $m_{j^*} \geq 0$ we have $m_{j^*} - r + 1 \geq 1 - r > -r$, while $\deg_\infty \Psi^{\mathrm{C},-} \leq -r$ by Lemma 5. The two orders are distinct, so $\deg_\infty(\Psi^{\mathrm{C},+} + \Psi^{\mathrm{C},-}) = m_{j^*} - r + 1$ and the leading coefficient is that of $\Psi^{\mathrm{C},+}$. A nonzero element of $K((w_n^{-1}))$ is nonzero. □

4 The reflection

Lemma 9 (reflection identity; P5)**.** *Let ι be the $\mathbb{Q}$-algebra automorphism of $\mathbb{Q}(w_1, \ldots, w_n)$ with $\iota(w_t) = w_t^{-1}$. Then for all $r \geq 0$ and $\theta \in \mathbb{Z}^r$,*

$$\iota\big(\Psi^{\mathrm{C}}_{(1^r),\theta}(M)\big) = (-1)^r e^{2r}\,\Psi^{\mathrm{C}}_{(1^r),\theta^\vee}(M), \qquad m_j^\vee = -m_j, \quad \textit{i.e.} \quad \theta_j^\vee = -\theta_j - 2j + 2.$$

Proof. Apply ι to Theorem 2 termwise; ι permutes nothing, so the index set $\{T\}$ and the index order within each T are unchanged, and $x_i \mapsto x_i^{-1}$. Three elementary identities:

$$1 - x^2 = -x^2\big(1 - x^{-2}\big), \qquad 1 - xy = -xy\Big(1 - \tfrac{1}{xy}\Big), \qquad \frac{1 - w_t x}{1 - \frac{x}{w_t}} = w_t^2\,\frac{1 - \frac{1}{w_t x}}{1 - \frac{w_t}{x}},$$

the last because $1 - w_t x = -w_t x\big(1 - \frac{1}{w_t x}\big)$ and $1 - \frac{x}{w_t} = -\frac{x}{w_t}\big(1 - \frac{w_t}{x}\big)$. Hence, using $\prod_{\{x,y\}\subseteq T} xy = \prod_{x\in T} x^{\,r-1}$ and $\prod_{t\notin T} w_t^{2r} = \big(e/\prod_{x\in T} x\big)^{2r}$,

$$\iota\big(B(T)\big) = (-1)^{r+\binom{r}{2}}\Big(\prod_{x\in T} x^2\Big)\Big(\prod_{x\in T} x^{\,r-1}\Big)\frac{e^{2r}}{\prod_{x\in T} x^{2r}}\,B(T) = (-1)^{r+\binom{r}{2}} e^{2r}\Big(\prod_{x\in T} x^{1-r}\Big)B(T).$$

For the alternant ratio, $\prod_{i<k}(x_k^{-1} - x_i^{-1}) = (-1)^{\binom{r}{2}} V(T)/\prod_i x_i^{r-1}$, so

$$\iota\left(\frac{\det(x_i^{m_j})}{V(T)}\right) = (-1)^{\binom{r}{2}}\frac{\big(\prod_i x_i^{r-1}\big)\det(x_i^{-m_j})}{V(T)} = (-1)^{\binom{r}{2}}\frac{\det\big(x_i^{r-1-m_j}\big)}{V(T)}.$$

Multiplying, the two signs $(-1)^{\binom{r}{2}}$ cancel and $\big(\prod_i x_i^{1-r}\big)\det(x_i^{r-1-m_j}) = \det(x_i^{-m_j})$, leaving $\iota(\Psi^{\mathrm{C}}_{(1^r),\theta}) = (-1)^r e^{2r}\sum_T B(T)\det(x_i^{-m_j})/V(T)$. By Theorem 2 this last sum is $\Psi^{\mathrm{C}}_{(1^r),\theta^\vee}(M)$ for the unique $\theta^\vee$ with ρ-shift $-m$, namely $\theta_j^\vee = -m_j - j + 1 = -\theta_j - 2j + 2$. □

Remark 10. For $r = 1$ this is the duality $I_a(1/w) = -I_{-2-a}(w)$ of `clio-Q553-reflection-duality-fixed-poin` in the normalisation $I_a = e\,\Psi^{\mathrm{C}}_{(1),a+1}$. Lemma 9 is its extension to all r, and the point that the reflection acts on the ρ-shift by the *clean* involution $m \mapsto -m$ — not by $m \mapsto 2r - 2 - m$, which is what I first derived and which the computation refuted — is what makes the case split of Theorem 11 close.

5 The theorem

Theorem 11 (Q559: the converse of the ρ-shift vanishing law)**.** *Let M be a set of n indeterminates, let $r \geq 0$ with $n \geq r$, and let $\theta \in \mathbb{Z}^r$ have pairwise distinct ρ-shifts $m_j = \theta_j + j - 1$. Then*

$$\Psi^{\mathrm{C}}_{(1^r),\theta}(M) \;\neq\; 0 \quad \text{in } \mathbb{Q}(w_1, \ldots, w_n).$$

Proof. Induction on r. For $r = 0$, $\Psi^{\mathrm{C}}_{(1^0),()}(M) = 1 \neq 0$.

Let $r \geq 1$ and assume the statement for $r - 1$, for every index set of size $\geq r - 1$ and every θ' with pairwise distinct ρ-shifts. Since $n \geq r$ we have $|M'| = n - 1 \geq r - 1$, and $\hat{m}$ — being a subfamily of m — is pairwise distinct, so the inductive hypothesis gives

$$\Psi^{\mathrm{C}}_{(1^{r-1}),\hat{\theta}}(M') \neq 0 \tag{3}$$

for $\hat{\theta}$ as in Lemma 7.

Case A: $\max_j m_j \geq 0$. Proposition 8 applies by (3), and gives $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) \neq 0$.

Case B: $\max_j m_j \leq -1$. Then $\min_j m_j \leq -1$, so the reflected shift $m^\vee = -m$ satisfies $\max_j m_j^\vee = -\min_j m_j \geq 1 \geq 0$ and is pairwise distinct. Case A applied to $\theta^\vee$ — legitimate, because Case A invokes only the inductive hypothesis at rank $r - 1$, which (3) supplies for $\theta^\vee$ as well — gives $\Psi^{\mathrm{C}}_{(1^r),\theta^\vee}(M) \neq 0$. By Lemma 9, $\iota\big(\Psi^{\mathrm{C}}_{(1^r),\theta}(M)\big) = (-1)^r e^{2r} \Psi^{\mathrm{C}}_{(1^r),\theta^\vee}(M) \neq 0$, since $e \neq 0$. As ι is a field automorphism it is injective, so $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) \neq 0$.

The two cases are exhaustive, which completes the induction. □

Corollary 12 (complete vanishing criterion)**.** *For M a set of n indeterminates, $r \geq 0$ and $\theta \in \mathbb{Z}^r$,*

$$\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0 \iff n < r \quad \text{or} \quad m_j = m_k \text{ for some } j < k.$$

Equivalently: the zero locus of $\theta \mapsto \Psi^{\mathrm{C}}_{(1^r),\theta}(M)$ is, for $n \geq r$, exactly the union of the $\binom{r}{2}$ hyperplanes $\theta_j - \theta_k = k - j$ in $\mathbb{Z}^r$.

Proof. "⇐" is Theorem 3 and Proposition 13; "⇒" is Theorem 11. □

Proposition 13 (the hypothesis $n \geq r$ is necessary)**.** *If $n < r$ then $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ for every θ, including every θ with pairwise distinct ρ-shifts.*

Proof. There is no injection h from r slots into an n-element set, so the sum (1) is empty. (Equivalently, M has no r-subset in Theorem 2.) □

This hypothesis was absent from Q559 as I posed it on 2026-10-22; the measured 424/424 sample never reached $n < r$, so the omission was invisible to the evidence that graded the conjecture. It costs nothing here, but it is the fourth time this month that a census has turned out to be a claim about a domain I had not written down.

Corollary 14 (one-step branching)**.** *Under the hypotheses of Theorem 11 with $\max_j m_j \geq 0$,*

$$\lim_{w_n \to \infty} w_n^{-(m_{j^*} - r + 1)}\, \Psi^{\mathrm{C}}_{(1^r),\theta}(M) = (-1)^{r+j^*}\, \Psi^{\mathrm{C}}_{(1^{r-1}),\hat{\theta}}\big(M \setminus \{w_n\}\big),$$

the limit taken coefficientwise in $K((w_n^{-1}))$. The operation "send the largest variable to infinity" lowers the rank by one and deletes the largest ρ-shift, *with a sign $(-1)^{r+j^*}$ recording its slot.*

Remark 15 (what the shape of this says)**.** Corollary 14 is the structural content of the proof, and it is prettier than the non-vanishing statement it was built for. The ρ-shift m is not merely a bookkeeping device that converts θ into exponents; it is the object on which the geometry of M acts. Removing a variable at infinity removes the *top* shift, so the multiset $\{m_j\}$ behaves like a set of occupied levels and $w_n \to \infty$ empties the highest one. That is why the vanishing locus is exactly the collision locus $m_j = m_k$: two particles in one level, and the configuration dies. The alternant was already telling me this — an alternant with a repeated exponent is two fermions in a state — but I had read it as a determinant trick rather than as a statement about levels.

6 Non-vanishing is generic, not universal

Theorem 11 is a statement in $\mathbb{Q}(w_1, \dots, w_n)$, i.e. about indeterminate M. At special points of M it fails, and the failure is structural rather than accidental.

Proposition 16 (reciprocal pairs; P3)**.** *Let $w_1, \dots, w_n$ be pairwise distinct nonzero complex numbers, so that every denominator in (1) and (2) is nonzero and both formulas evaluate. Suppose $w_a w_b = 1$ for some $a \neq b$. If $n \leq r+1$ then $B(T) = 0$ for* every *r-subset T, hence $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ for* every *θ — including every θ with pairwise distinct ρ-shifts.*

Proof. Let $|T| = r$. Since $|M \setminus T| = n - r \leq 1$, T cannot omit both a and b. If $a, b \in T$ then the pair factor $1 - \frac{1}{w_a w_b} = 0$ in (2). If exactly one of them lies in T, say $a \in T$ and $b \in M \setminus T$, then the factor $1 - \frac{1}{w_b w_a}$ of the third product of (2) (with $t = b$, $x = w_a$) is 0. Either way $B(T) = 0$, and Theorem 2 is a sum of terms each containing $B(T)$. □

Conversely, when $n \geq r + 2$ there exists an r-subset avoiding both a and b, and the conclusion fails: measured in §8, the vanishing is total for $n \leq r + 1$ and absent for $n \geq r + 2$, in 8 out of 8 cells. For $r = 1$, Proposition 16 recovers a fact the residue picture explains: at $n = 2$ with $w_2 = 1/w_1$ the kernel $H(z) = \prod_t \frac{zw_t - 1}{z - w_t}$ degenerates to $H \equiv 1$, so every coefficient h_k $(k \geq 1)$ and $\tilde{h}_a$ $(a \geq 1)$ vanishes and the three-case closed form returns 0 in all three cases.

Proposition 17 (self-reciprocal points; the $a = b$ degeneration)**.** *Let $w_1, \dots, w_n$ be pairwise distinct nonzero complex numbers and suppose $w_a^2 = 1$ — that is, $w_a = \pm 1$ — for some single index a. If $n \leq r$ then $B(T) = 0$ for* every *r-subset $T \subseteq M$, hence $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ for* every *θ.*

Proof. Let $T \subseteq M$ with $|T| = r$. Then $r = |T| \leq |M| = n \leq r$, so $T = M$ and in particular $a \in T$. The factor $1 - w_a^{-2}$ of the *first* product of (2) is therefore 0, and Theorem 2 is a sum of terms each containing $B(T)$. □

Two warnings about how this sits next to Proposition 16, both of which cost me a wrong sentence in the first version of this note.

The threshold is different. Proposition 16 needs only that T *meet* $\{a, b\}$, which holds as soon as $n \leq r + 1$; here T must *contain* the single index a, which needs $n \leq r$. So the self-reciprocal arm is strictly narrower, and it is non-vacuous only at $n = r$ (for $n < r$ there are no r-subsets at all). A threshold that shifts by one is not a cosmetic variant of the same statement.

The vanishing factor is a different factor. It is tempting to read $w_a = \pm 1$ as the case $b = a$ of $w_a w_b = 1$ and conclude that the pair factor $1 - \frac{1}{xy}$ of (2) degenerates. It does not: $\{a, a\}$ is the singleton $\{a\}$, which is not a two-element subset of T, so the second product of (2) never sees the degeneration. The vanishing happens in the *first* product, at the diagonal factor $1 - x^{-2}$. The

unification of the two conditions is real, but only over a *multiset* of indices, where $\{a, a\}$ keeps its multiplicity and $1 - \frac{1}{w_a w_a} = 1 - w_a^{-2}$; over sets it is false.

Both arms, both thresholds, and both of these warnings are machine-checked: the module `TworowD4Kernel.ReciprocalPair` of `lean/tworow_d4_kernel` proves Proposition 16 as `Bfactor_eq_zero_of_rec` and Proposition 17 as `Bfactor_eq_zero_of_self_reciprocal`, sorry-free on the three standard axioms, together with a refutation (`L2_distinctness_irremovable_at_succ_threshold`, witness $w = (1, 2)$, $r = 1$, $n = 2$, $B(\{1\}) = \frac{3}{4} \neq 0$) showing that Proposition 17 is *false* at Proposition 16's threshold $n \leq r+1$, and the set-versus-multiset separation `pairFactor_singleton_ne_diagFactor`.

Remark 18 (how I found this)**.** This witness came out of a negative control that read 58/60 instead of 60/60. The control was the no-shift alternant (using $m_j = \theta_j$ in place of $\theta_j + j - 1$), which must *disagree* with the definition. Two of the sixty agreed. One was an expected degeneracy — both θ and m had a repetition, so both sides were 0. The other was $r = 2$, $n = 3$, $\theta = (-1, 3)$, $m = (-1, 4)$ pairwise distinct, value 0: for a moment, a counterexample to the theorem of this note. It was $w = (6, \frac{5}{3}, \frac{3}{5})$, and $\frac{5}{3} \cdot \frac{3}{5} = 1$. **A half-dead control is a measurement**, for the third time this month; this time what it measured was a whole proposition.

7 Grading the pre-registration

P1 — `SPLIT`**.** The predicted arm, and the prediction's three reasons all stood.

(a) The brief's strategy-1 is indeed ill-posed as literally stated: the summands $B(T) \det / V$ are not polynomials in w, so there is no "monomial occurring in exactly one T-summand" to find. I did not pursue a repair by clearing denominators; the common denominator mixes the T's, which is exactly the structure the argument wanted to keep separate.

(b) A leading-term argument does close the problem, in the grading by order at $w_n \to \infty$ over K rather than by a monomial order on all of w.

(c) It closes exactly the class

$$\Big\{\theta : \max_j m_j \geq 0\Big\} \cup \Big\{\theta : \max_j m_j = -k \leq -2 \text{ and } n \leq r + k - 1\Big\},$$

and no more. The second family is where one may iterate Lemma 5 instead: when $\max_j m_j \leq -2$ the block $\Psi^{\mathrm{C},-}$ strictly dominates, with leading coefficient $\pm\Psi^{\mathrm{C}}_{(1^r),\theta+\mathbf{1}}(M')$, so one descends in n while $\max_j m_j$ rises by one each step, reaching the direct base case $n = r$ (where $\Psi^{\mathrm{C},-}$ is empty and $\Psi^{\mathrm{C}} = B(M) \det / V \neq 0$) after $n - r$ steps — provided $\max_j m_j$ has not meanwhile arrived at -1. It arrives at -1 after $k - 1$ steps, so the descent wins iff $n - r \leq k - 1$.

The residual class is therefore nonempty: $\max_j m_j = -1$ with $n > r$, and $\max_j m_j = -k \leq -2$ with $n \geq r + k$. On it the two blocks have *equal* order $-r$ — measured to be an exact order tie in 25/25 instances with $\max_j m_j = -1$ — so the leading coefficient of Ψ^{C} is the *sum* $c_+ + c_-$ of two separately nonzero quantities and the instrument has no purchase. Empirically $c_+ + c_- \neq 0$ in all 25, so the residual class is not where the theorem fails; it is where this argument fails. The reflection of Lemma 9 is therefore **essential to the proof, not a convenience**: it maps the residual class into Case A by $m \mapsto -m$.

P2 — `TRUE`**.** $B(T)$ takes both signs in 400/400 sampled $r = 2$ instances with $n \in \{3, 4\}$ and real positive w. Witness $w = (\frac{9}{4}, \frac{1}{4}, \frac{3}{2}, \frac{5}{2})$: $B(\{1, 2\}) = -\frac{63973}{19683}$ and $B(\{1, 3\}) = \frac{703703}{157464}$. So the sign problem of §2.1 is real and not an artefact of my framing; the proof does not dissolve it but routes around it, by never summing the terms at all — the two blocks are separated by *order*, not by sign.

P3 — `CONFIRMED`**, with the structural prediction also confirmed.** Proposition 16: non-vanishing holds generically (Theorem 11) and fails on the reciprocal-pair locus when $n \le r+1$. The predicted threshold is exact: §8 measures total vanishing in all cells with $n \le r+1$ and none in any cell with $n \ge r+2$. The distinction the brief asked me to keep — identically zero as a rational function versus zero at a sample point — is exactly the distinction between Corollary 12 and Proposition 16.

P4 — `CONFIRMED` , both exponents and both signs, 137/137 on each block separately, with three live negative controls at 0/52.

P5 — `CONFIRMED` , 300/300, verified against the head-injection *definition* (1) rather than against the alternant form used to derive it.

8 Verification

All arithmetic is exact: Python `fractions.Fraction` for numerics, SymPy for the rational-function work, using `cancel` only — `nsimplify` is barred, having manufactured $2^{41/58}$ from a rational twice this month. Two independent implementations of Ψ^{C} were written: `psi_headsum` from the definition (1) (a sum over ordered injections) and `psi_alt` from Theorem 2 (a sum over subsets). Scripts and the verbatim pre-registration are published with this note at `code-1023-q559/` in the same repository: `PREREG.md`, `psi.py` (both implementations), `sampler.py`, `t1`–`t8*.py`. They were written in `scratch/`, which is not a published repository; a cited script that nobody can open is the same fault as an answer written to the wrong artifact class.

Sampler hygiene. After Propositions 16 and 17 the sampler excludes $w_a = \pm 1$, $w_a w_b = 1$ and $w_a = w_b$. The reciprocal-pair exclusion is not cosmetic: it is a genuine zero locus of Ψ^{C}, and sampling across it manufactures false counterexamples. *Correction.* The first version of this note justified all three exclusions by Proposition 16 alone. That proposition assumes $a \neq b$ and therefore says nothing about $w_a = \pm 1$; the warrant for that arm is Proposition 17, at its own threshold $n \le r$. The exclusion was sound and the citation was wrong, which is a weaker defect than an unsound exclusion and a worse one than a missing citation: an exclusion cited to a proposition that does not cover it reads as proved and is not.

Claim	Range	Result
Thm 2, definition vs. alternant	$r \le 3$, $r \le n \le r+2$	386/386
Thm 2, again on the sweep sampler	$r \in \{2,3\}$, $n \le r+2$	120/120
Lem 9 (P5) vs. the definition	$r \le 3$, $r \le n \le r+2$	300/300
Lem 7 (P4, block $T \ni n$): exponent *and* coefficient	$r \le 3$, $n \in \{r{+}1, r{+}2\}$	137/137
Lem 5 (P4, block $T \not\ni n$): exponent *and* coefficient	$r \le 3$, $n \in \{r{+}1, r{+}2\}$	137/137
Lem 7(i) $\lim B_M(T' \sqcup \{n\}) = B_{M'}(T')$	$r \le 3$, $n \in \{r{+}1, r{+}2\}$	204/204
Lem 5 step $\lim w_n^r B_M(T) = (-1)^r(\prod x) B_{M'}(T)$	idem	152/152
row scaling $(\prod x)\det(x^m) = \det(x^{m+1})$	idem	152/152
Prop 16 threshold, zero iff $n \le r+1$	$r \le 3$, $n \in \{r, r{+}1, r{+}2\}$	8/8 cells
Prop 13, $n < r \Rightarrow \Psi^{\mathrm{C}} = 0$	$r \le 4$, $n < r$, distinct m	911/911
Order tie at $\max_j m_j = -1$: $\deg_\infty \Psi^{\mathrm{C},+} = \deg_\infty \Psi^{\mathrm{C},-} = -r$	$r \in \{2,3\}$	25/25
and in those, $\mathrm{lc}_\infty \Psi^{\mathrm{C},+} + \mathrm{lc}_\infty \Psi^{\mathrm{C},-} \ne 0$	idem	25/25
Thm 11, refutation arm	$r{=}2$, $\theta \in [-5,5]^2$, $n \le 5$	0 zeros / 888
	$r{=}3$, $\theta \in [-4,4]^3$, $n \le 5$	0 zeros / 3216
	$r{=}4$, $\theta \in [-3,3]^4$, $n \le 5$	0 zeros / 4584
	total, distinct m, two w per cell	0 zeros / 8688

8.1 Controls

Negative control	What it breaks	False-pass rate
no-shift alternant ($m_j = \theta_j$)	Thm 2	2/60, both diagnosed
e^2 in place of e^{2r}	Lem 9	6/60, all diagnosed
$\theta^\vee = \theta$	Lem 9	9/60, all diagnosed
non-argmax j in place of j^*	Lem 7	0/52
sign $(-1)^{r+j^*}$ dropped	Lem 7	36/52, see below
θ in place of $\theta + \mathbf{1}$	Lem 5	0/52
exponent $-(r{-}1)$ in place of $-r$	Lem 5	0/52
Positive control		
repeated $m \Rightarrow \Psi^{\mathrm{C}} = 0$ on the sweep instrument	Thm 3	2861/2861

Every control that was not fully live has been diagnosed rather than accepted.

- *The three half-dead controls on Theorem 2 and Lemma 9* (2, 6 and 9 hits) are **entirely** the repeated-m degeneracy: in all 17 cases both sides were 0 by Theorem 3 (note $-m$ has a repetition iff m does, which is why the reflection controls inherit it). The single exception, the 58/60 hit with distinct m, is Remark 18's reciprocal pair and became Proposition 16. Restricted to distinct m and a hygienic sampler, all three controls are fully live.

- *The* 36/52 *control* (dropping the sign $(-1)^{r+j^*}$) is live only on the subfamily where that sign is -1. Counted directly: the sign was $+1$ in 36 of the 52 cases and -1 in 16, and the control disagreed in exactly those 16, i.e. 16/16 where it has any power to disagree. It is a partial control by construction, and I report it as such rather than as a pass.

9 What is proved, what is not

- **Proved.** Theorem 11 and hence Corollary 12: a complete vanishing criterion for $\Psi^{\mathrm{C}}_{(1^r),\theta}(M)$ over indeterminate M. Lemmas 5, 7, 9, Propositions 8, 13, 16, Corollary 14. Every step is

an identity in $K((w_n^{-1}))$ or in $\mathbb{Q}(w)$; no genericity, no specialisation, no numerical limit is used in any proof.

- **Standing on.** Theorem 2, proved 2026-10-22 and graded `proved` (`clio-Q553-alternant-expansion`), re-verified here against the definition 386/386. Nothing else external: there are no citations in this note, and so no locator obligations.

- **Not proved, and not needed.** That $\mathrm{lc}_\infty \Psi^{\mathrm{C},+} + \mathrm{lc}_\infty \Psi^{\mathrm{C},-} \neq 0$ on the residual class $\max_j m_j = -1$. Measured 25/25, graded `computed`. The theorem does not rest on it — the reflection bypasses the class entirely — so this is an open question about the *instrument*, not a gap in the proof. If it is true in general it would give a second, reflection-free proof.

- **Open (new).** Is the union of the two structural loci the *whole* zero locus of $\Psi^{\mathrm{C}}_{(1^r),\theta}$ as a function of M at fixed distinct θ? The question must be asked with both thresholds carried, which the first version of this note did not do: the reciprocal-pair arm $w_a w_b = 1$, $a \neq b$ forces vanishing for $n \leq r+1$ (Proposition 16), while the self-reciprocal arm $w_a = \pm 1$ does so only for $n \leq r$ (Proposition 17). Writing the candidate locus as "the reciprocal-pair locus together with $w_a = \pm 1$" suggests one proposition delivering both arms at one threshold, and no such proposition is proved here. For $r = 1$, $n = 2$ the answer is yes; the general shape of the locus is unmeasured here.